Yuccas and their moths



In the hot, arid regions of the Americas, there is a remarkable mutualistic relationship between yucca shrubs and yucca moths. No other insects pollinate these plants, and no other plants host yucca moth caterpillars. A female yucca moth collects pollen from the flower of one yucca plant and deposits it in the flower of another yucca, fertilizing the plant as it does so. The moth then cuts a hole in the flower's ovary and lays an egg; she may lay several in the same flower. When the eggs hatch, the caterpillars feed on the seeds developing in the flower but do not eat them all, leaving enough for the plant to propagate. If too many eggs are laid in one flower, the plant sheds it before the caterpillars hatch – leaving those insects to starve. Without these moths, the yuccas would not pollinate and would soon die out. Without the yuccas, the moths would have nowhere to lay and nurture their eggs, and they too would not survive.

See also: Evolution by natural selection • Ecological niches • Competitive exclusion principle • Animal ecology

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